

Math 4101 Practice Midterm
Extra Practice Set

Name: _____

Time: 50 minutes

Total: 100 points

Directions. Write complete proofs or arguments. Clearly state any definitions you use. Unless otherwise stated, all numbers are real. Throughout this exam, $\mathbb{N} = \{1, 2, 3, \dots\}$.

Problem 1 (30 points).

Let

$$S = \{x \in \mathbb{R} : x^3 \in \mathbb{Q}\}.$$

- (a) (15 points) Prove that S is countably infinite.
- (b) (15 points) Prove that $\mathbb{R} \setminus S$ is uncountable.

Problem 2 (30 points).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and write $m = \inf A$.

- (a) (15 points) Prove that for every $\varepsilon > 0$, there exists $a \in A$ such that

$$m \leq a < m + \varepsilon.$$

- (b) (15 points) Define

$$T = \left\{ a + \frac{1}{n} : a \in A, n \in \mathbb{N} \right\}.$$

Prove that $\inf T = m$. You may use part (a), even if you have not proved it.

Problem 3 (40 points).

Use the standard metric $d(x, y) = |x - y|$ on $\mathbb{R}$.

- (a) (20 points) Let $U, V \subseteq \mathbb{R}$ be open. Prove directly from the definition of openness that $U \cup V$ is open.
- (b) (20 points) Suppose $U_n \subseteq \mathbb{R}$ is open for every $n \in \mathbb{N}$. Must

$$\bigcup_{n=1}^{\infty} U_n$$

be open? Give a proof or an explicit counterexample. If you give a counterexample, determine the union and justify your answer.

Problem 4 (Bonus, 10 points).

Suppose $F_n \subseteq \mathbb{R}$ is closed for every $n \in \mathbb{N}$. Must

$$\bigcup_{n=1}^{\infty} F_n$$

be closed? Give a proof or an explicit counterexample. (Hint: think about how this problem relates to Problem 3, part (b), via complements and De Morgan's laws.)